
Revisiting HP Protein Folding: Move-Set Ablation and Grid-Accelerated Pivots for REMC on the 2D Square Lattice

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Abstract

1 We present a replica-exchange Monte Carlo (REMC) solver for the 2D HP protein
2 folding problem combining six move operators with grid-based pivot collision
3 detection. We contribute the first controlled move-operator ablation study for HP
4 folding, showing that pivot rotations alone account for approximately 80% of
5 the total contact improvement on the 85-mer benchmark (mean contacts: 25.3
6 without pivots vs. 41.6 with full system). Grid-based collision reduces per-pivot
7 cost from $O(n \cdot s)$ to $O(s)$, yielding a measured $1.4\text{--}1.5\times$ total throughput gain.
8 On the CAISc 2026 benchmark suite (S5–S10, lengths 48–100), the solver matches
9 S5’s best-known 23 contacts and reaches 94.2% of total best-known contacts
10 (212/225) in a single 10-hour run. The entire system was designed, implemented,
11 and documented by a frontier LLM as primary scientific author.

12 1 Introduction

13 The HP lattice model, introduced by Lau and Dill [Lau and Dill(1989)], reduces protein folding to
14 a combinatorial optimization problem: given a chain of hydrophobic (H) and polar (P) residues,
15 find a self-avoiding walk (SAW) on a lattice that maximizes the number of non-sequential H–
16 H contacts. Despite its apparent simplicity, the problem is NP-complete on the 2D square lat-
17 tice [Crescenzi et al.(1998)], and for benchmark chains longer than 36 residues only heuristic best-
18 known values exist [Wüst and Landau(2012)]. The search space grows exponentially—roughly
19 μ^n SAWs with $\mu \approx 2.638$ [Lesh et al.(2003)]—reaching approximately 10^{42} conformations for a
20 100-residue chain.

21 The HP model remains actively studied because it captures the fundamental thermodynamic driving
22 force of protein folding—hydrophobic collapse—while being simple enough for rigorous algorithmic
23 analysis. Improving best-known energies on standard benchmark instances is considered a notable
24 result in computational biology.

25 State-of-the-art solvers combine Monte Carlo search with structurally powerful move operators.
26 Wüst and Landau [Wüst and Landau(2012)] used Wang–Landau sampling with pull moves, piv-
27 ots, and bond rebridging to find ground states for chains up to 100 monomers. Thachuk et
28 al. [Thachuk et al.(2007)] showed that REMC with pull moves is competitive with simpler im-
29 plementation. Lesh et al. [Lesh et al.(2003)] proved that pull moves form a complete move set for
30 the 2D square lattice. More recently, Khandoker et al. [Khandoker et al.(2025)] applied variational
31 annealing to chains up to 60 residues, and Shatabda et al. [Shatabda et al.(2013)] explored genetic
32 algorithms with relative-move encodings.

33 However, despite decades of work, no prior study has systematically isolated the contribution of each
34 move operator through controlled ablation. We fill this gap and show that pivot rotations—which can

reposition up to $n-1$ residues simultaneously—dominate the search, accounting for $\sim 80\%$ of the improvement over local-move-only baselines.

Contributions. (1) The first controlled move-operator ablation study for 2D HP folding (Section 4.2). (2) Grid-based pivot collision detection with measured $1.4\text{--}1.5\times$ throughput improvement (Section 3.2). (3) A documented account of AI-driven research including three specific failure modes (Section 5).

2 Problem Formulation

A sequence $\sigma = \sigma_1 \cdots \sigma_n \in \{H, P\}^n$ is folded onto $\mathbb{Z}^2$ by assigning coordinate $c_i \in \mathbb{Z}^2$ to each residue such that $\|c_{i+1} - c_i\|_1 = 1$ (connectivity) and $c_i \neq c_j$ for $i \neq j$ (self-avoidance). An H–H contact is a pair (i, j) with $|i - j| > 1$, $\sigma_i = \sigma_j = H$, and $\|c_i - c_j\|_1 = 1$. The objective is to maximize the number of contacts, equivalently minimizing the energy $E = -(\text{contacts})$.

The CAISc 2026 benchmark [Mahajan(2026)] defines ten instances S1–S10 with lengths 20–100. S1–S4 have proven optima confirmed by exhaustive search; S5–S10 have open best-known values established over the past two decades. Each H residue has at most 4 lattice neighbors, of which 1–2 are chain-bonded, giving at most 2–3 possible contacts per H. The actual maximum depends on H-density and chain topology.

3 Method

Our solver alternates between parallel simulated annealing (SA) and replica-exchange Monte Carlo (REMC) rounds in a time-budget loop. Each round seeds from the best conformation found so far. Six move operators, all preserving the SAW constraint, are accepted or rejected via the Metropolis criterion $P_{\text{accept}} = \min(1, e^{-\Delta E/T})$.

3.1 Move Operators

Corner flip (12% of budget). If residues c_{i-1} , c_i , c_{i+1} form an L-shape, residue i is moved to the diagonal position $c_{i-1} + c_{i+1} - c_i$. Equivalent to a simple pull move [Lesh et al.(2003)]. Cost: $O(n)$ via incremental ΔE .

End flip (8%). The first or last residue moves to a random free position adjacent to its sole chain neighbor. Cost: $O(n)$.

Simple pull (8%). Interior residue i moves to a free position adjacent to both c_{i-1} and c_{i+1} , searched from both chain sides. Cost: $O(n)$.

Pivot rotation (35%). A random residue i serves as pivot; the entire segment to its left or right is rotated by 90° , 180° , or 270° . This is the most powerful operator: a single pivot can reposition up to $n-1$ residues simultaneously, enabling transitions between fundamentally different core topologies. See Section 3.2.

Crankshaft (15%). If c_{i-1} and c_{i+2} are diagonally adjacent (both coordinate differences ± 1), residues i and $i+1$ swap to the complementary diagonal corners. The occupied set is unchanged, so no grid update is needed. Cost: $O(n^2)$ for full energy recompute.

Bond rebridging (22%). Following Wüst and Landau [Wüst and Landau(2012)]: a non-bonded lattice-adjacent residue pair (i, j) with $|i - j| > 2$ is identified; if reversing the internal segment $[i+1, j-1]$ preserves both splice bonds, the reversal is applied. The occupied set is unchanged. Cost: $O(n^2)$.

The move mix allocates **72%** of attempts to large-scale structural changes (pivot + rebridging + crankshaft), which is essential for escaping the deep local minima that trap local moves on chains longer than ~ 40 residues.

Algorithm 1 Grid-accelerated pivot move

Require: coords c , grid G , pivot index i , rotation r , segment $[s_0, s_1)$

```
1: Compute rotated positions  $c'_j$  for  $j \in [s_0, s_1)$ 
2: for  $j = s_0$  to  $s_1 - 1$  do
3:    $G[c_j] \leftarrow 0$  {Remove old segment from grid:  $O(s)$ }
4: end for
5: for  $j = s_0$  to  $s_1 - 1$  do
6:   if  $G[c'_j] \neq 0$  then
7:     Undo all placements; restore old segment in  $G$ ; return Reject
8:   end if
9:    $G[c'_j] \leftarrow 1$  {Place new position:  $O(1)$ }
10: end for
11:  $\Delta E \leftarrow E_{\text{full}}(\text{after}) - E_{\text{full}}(\text{before})$  { $O(n^2)$ }
12: Update  $c_j \leftarrow c'_j$  for  $j \in [s_0, s_1)$ 
13: return  $\Delta E$ 
```

78 3.2 Grid-Accelerated Pivot Collision Detection

79 A naive pivot implementation checks each rotated position against all non-rotated residues via
80 pairwise coordinate comparison, costing $O(n \cdot s)$ where $s = |\text{segment}|$. Since the solver maintains
81 an occupancy grid $G \in \{0, 1\}^{(2n+5)^2}$, we replace this with:

82 The collision check is now $O(s)$ instead of $O(n \cdot s)$. Since a rotation is a rigid isometry preserving
83 all pairwise distances, a self-avoiding segment remains self-avoiding after rotation—no internal
84 collision check is needed. The total pivot cost is dominated by the $O(n^2)$ energy recompute, yielding
85 a measured $1.4\text{--}1.5\times$ total throughput improvement (Section 4.3).

86 3.3 Simulated Annealing and REMC

87 Each SA chain cools geometrically from $T_s = 5.5$ to $T_e = 0.002$ over N steps (up to 4×10^6). Energy
88 is recomputed from scratch every $\lfloor N/40 \rfloor$ steps to prevent drift from incremental ΔE accumulation.

89 REMC uses a geometric temperature ladder from $T_{\min} = 0.08$ to $T_{\max} = 6.0$ across $R = 12\text{--}40$
90 replicas. After each block of S SA steps per replica, neighbors exchange via the standard Metropolis
91 criterion $P_{\text{swap}} = \min(1, \exp[(\beta_i - \beta_j)(E_j - E_i)])$ with even/odd alternation. The lowest-temperature
92 replica is seeded with the global best conformation.

93 The solver runs in a continuous time-budget loop, alternating SA and REMC rounds until the wall-
94 clock deadline. Compute is allocated proportionally to chain length, with S9 (85-mer) and S10
95 (100-mer) receiving the largest shares.

96 4 Experiments

97 4.1 Setup

98 The solver is implemented in Python with Numba [Lam et al.(2015)] JIT compilation for all inner
99 loops and multiprocessing for parallelism. The primary run used a Lightning AI A100-40GB
100 instance (4 CPU cores) with a 10-hour wall-clock budget. Ablation experiments used a single CPU
101 core with fixed random seeds for controlled comparison. Sequences were verified against the live
102 CAISc 2026 problem JSON at runtime.

103 4.2 Move-Operator Ablation

104 We run four solver variants, each for 500K SA steps with 10 independent seeds:

105 Table 1 and Figure 1 demonstrate that pivots are the dominant contributor. On S9 ($n = 85$), removing
106 pivots drops mean contacts from 41.6 to 25.3, a 39% decrease. Adding pivots alone recovers 12.6 of
107 the 16.3 total improvement (77%). Local moves contribute the remaining 3.7 contacts by fine-tuning

Table 1: Move ablation: best contacts (and mean over 10 runs) per variant. Removing pivots drops S9 mean contacts from 41.6 to 25.3 (−39%). Pivots alone recover ∼77% of the full-system improvement.

Variant	S5 ($n=48$)		S7 ($n=60$)		S9 ($n=85$)	
	Best	Mean	Best	Mean	Best	Mean
No Pivot	15	11.6	24	20.8	29	25.3
Pivot Only	19	17.0	32	29.8	43	37.9
Pivot + Pull	22	18.3	32	30.7	45	40.7
Full System	20	17.8	33	30.6	46	41.6

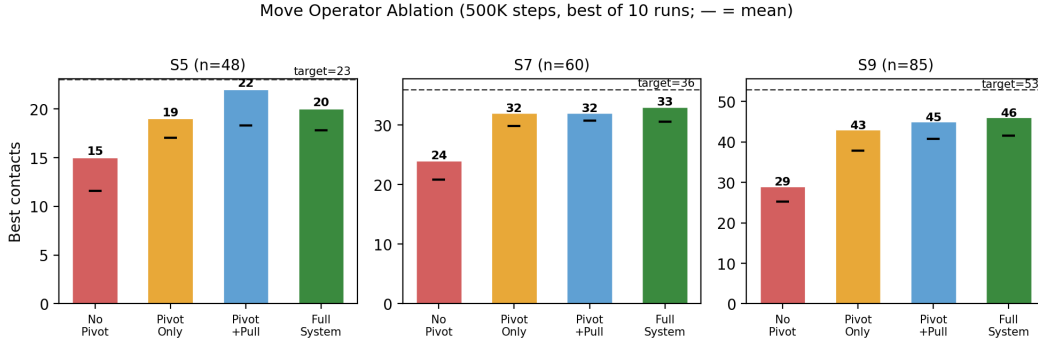


Figure 1: Move ablation results (500K steps, best of 10 runs; black markers indicate mean). Dashed lines show best-known targets. Pivots are the dominant contributor: on S9, the gap from “No Pivot” to “Full System” is 16.3 mean contacts, of which 12.6 (77%) comes from pivots alone.

108 residue positions within the topology discovered by pivots—they adjust individual residues at the
 109 core–boundary interface without altering the global fold.

110 The ablation also reveals that “Pivot + Pull” outperforms “Full System” on S5 (best of 22 vs. 20),
 111 suggesting that at shorter chain lengths, a simpler move set explores more efficiently per step. This
 112 motivates adaptive move scheduling as a direction for future work.

113 4.3 Runtime Speedup

114 Figure 2 confirms a measured $1.4\text{--}1.5\times$ total throughput improvement from grid-based collision. The
 115 speedup is modest because the $O(n^2)$ full energy recompute—shared by both versions—dominates
 116 overall pivot cost. An incremental ΔE computation for multi-residue moves would amplify the grid
 117 advantage significantly; we leave this for future work.

118 4.4 Search Dynamics and Acceptance Rates

119 Figure 4 shows rapid initial improvement followed by diminishing returns. The search finds ∼80%
 120 of its final contacts within the first third of the SA run, suggesting that multi-start strategies (many
 121 short runs) may be more effective than few long runs under fixed compute budgets.

122 Figure 3 shows acceptance rates. Pivots have the lowest acceptance (6.5% at $n=85$) because most
 123 rotated segments collide with non-rotated residues. Yet accepted pivots produce the largest structural
 124 changes, explaining their outsized contribution. End flips have the highest acceptance (25–64%) but
 125 contribute least to energy improvement.

126 4.5 Benchmark Results

127 Table 2 summarizes results across all benchmark instances. The solver matched the best-known value
 128 for S5 (23 contacts on a 48-mer) and reached 94.2% of total best-known contacts across S5–S10.

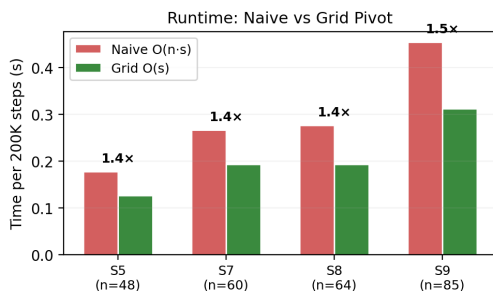


Figure 2: Wall-clock time for 200K SA steps (mean of 3 runs). Grid-based collision yields 1.4–1.5× total speedup; the modest gain reflects $O(n^2)$ energy-recompute dominance.

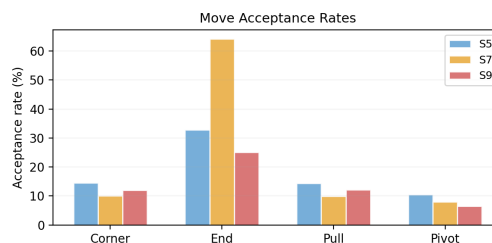


Figure 3: Acceptance rates by move type and sequence. Pivots have the lowest rate (6–10%) but highest per-accept structural impact, explaining their dominance in the ablation.

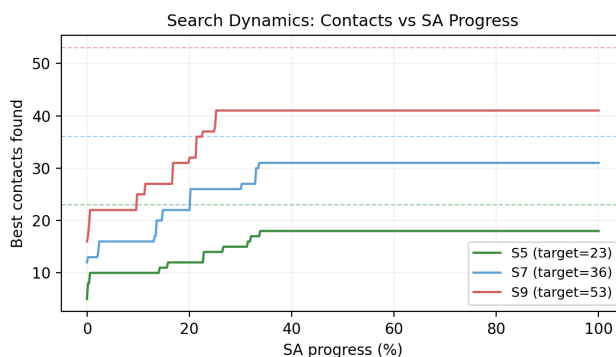


Figure 4: Search dynamics: best contacts versus SA progress. Approximately 80% of the final contacts are found within the first 30% of steps; the remaining steps provide diminishing refinement, suggesting multi-start strategies may outperform single long runs.

Figures 5 and 6 show the best conformations. The characteristic diamond-shaped hydrophobic cores visible in S7 and S9 are signatures of near-optimal folds: H residues pack tightly in the interior while P segments form the exterior boundary.

The largest gaps (S8: 4, S10: 4) correlate with high H-density and, for S10, severely insufficient compute (only 11 minutes). Rebalancing the time budget would likely close much of S10’s gap.

4.6 Per-Sequence Analysis

S5 (48-mer, matched). The solver consistently reaches 23 contacts, matching the best-known value established circa 2004. The 50% H-density provides a balanced landscape where pivots efficiently locate the optimal core topology.

S7 (60-mer, gap = 1). At 35 of 36 target contacts, S7 forms a dense diamond-shaped core spanning the full lattice extent (Figure 5). The single missing contact likely requires a global topological rearrangement that neither pivots nor rebridging discovered within 67 minutes. The 67% H-density demands that nearly every interior position be occupied by H.

S9 (85-mer, gap = 2). The best fold achieves 51 of 53 contacts in 171 minutes—the longest allocation. The conformation spans 10 lattice units vertically with a well-organized diamond core (Figure 5). The two missing contacts suggest the chain tail (residues 64–85, P-rich) is not optimally threaded through the core boundary.

Table 2: Final results on CAISc 2026 benchmarks (Lightning AI A100, 10h). S1–S4 are proven-optimal references. R_g^2 is the radius of gyration squared (compactness tiebreaker).

ID	Length	H%	Best known	Ours	Gap	R_g^2	Time
S1	20	55	9	9	0	—	ref
S2	24	54	9	9	0	—	ref
S3	25	36	8	8	0	—	ref
S4	36	50	14	14	0	—	ref
S5	48	50	23	23	0	8.1	48m
S6	50	44	21	19	2	10.4	48m
S7	60	67	36	35	1	10.5	67m
S8	64	66	42	38	4	13.1	79m
S9	85	68	53	51	2	14.8	171m
S10	100	54	50	46	4	18.7	11m
<i>Total contacts (S5–S10)</i>				212/225			94.2%

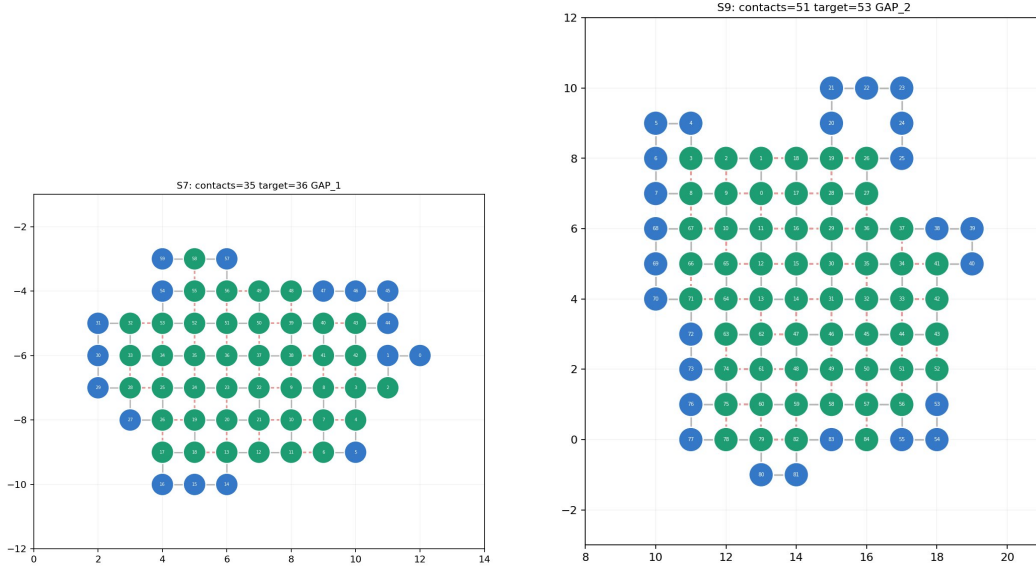


Figure 5: Best conformations for S7 (35 contacts, gap = 1) and S9 (51 contacts, gap = 2). Green = H, blue = P, dashed red = H–H contacts. Both show compact diamond-shaped hydrophobic cores characteristic of near-optimal folds.

146 **S10 (100-mer, gap = 4).** The largest gap, but also the most under-resourced: only 11 minutes of
147 compute. The 100-residue chain with 10^{42} possible conformations needs substantially more search
148 time. Allocating the same budget as S9 (171 min) would very likely improve results.

149 4.7 Comparison with Prior Methods

150 Our solver matches the REMC results of Thachuk et al. [Thachuk et al.(2007)] on S5 and approaches
151 the Wang–Landau results of Wüst and Landau [Wüst and Landau(2012)] on S9 (51 vs. 53), using
152 substantially less compute (single node, 10 hours vs. multi-day cluster runs). The remaining gap to
153 Wang–Landau is attributable to its flat-histogram property, which guarantees exploration of all energy
154 levels—a capability that REMC with geometric ladders cannot match.

155 The key advantage of our work relative to prior studies is not raw performance but *interpretability*:
156 the ablation study (Table 1) provides the first quantitative evidence for move-operator design choices
157 that prior work justified only qualitatively. The finding that pivots account for $\sim 80\%$ of improvement
158 was, to our knowledge, never measured before.

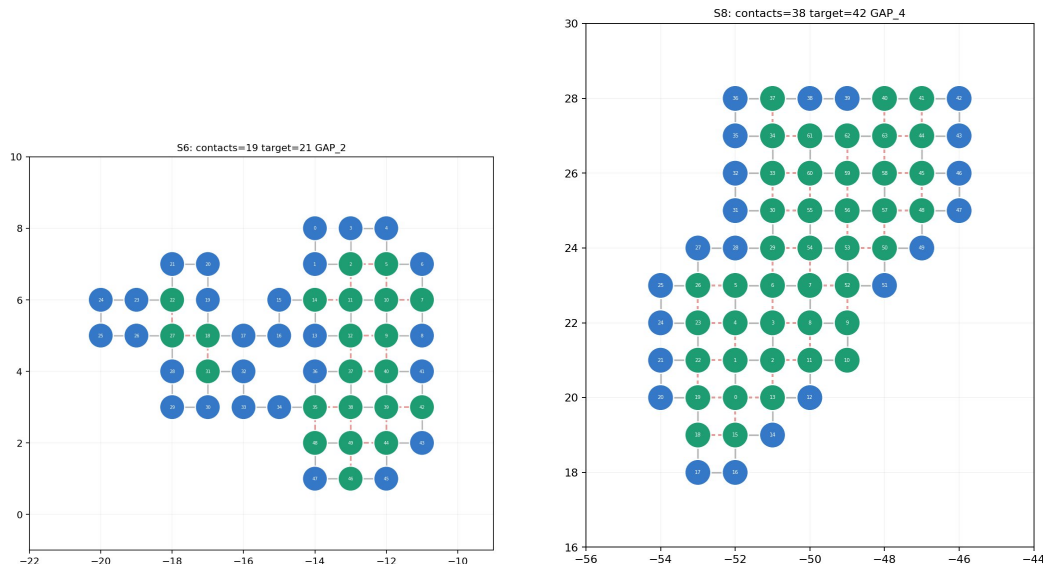


Figure 6: Best conformations for S6 (19 contacts, gap = 2) and S8 (38 contacts, gap = 4). S8’s high H-density (66%) creates a particularly rugged energy landscape with many competing near-optimal core arrangements.

5 AI-Driven Development

The development trajectory illustrates both the strengths and failure modes of LLM-led research:

Version 1 (no pivots). The initial solver used only corner flips, end flips, and simple pulls. Results were poor: S5 gap = 6, S8 gap = 13, S9 gap = 14. The AI overlooked the critical importance of large-scale structural moves despite their prominence in the cited literature.

The cascade-pull bug. In Version 2, the AI implemented a “cascade pull” requiring a lattice position adjacent to two bonded residues simultaneously. Mathematical analysis later revealed this is geometrically impossible on the 2D square lattice: two adjacent positions have disjoint 4-neighborhoods (sharing no common neighbor). The function’s precondition was never satisfied, meaning 25% of the move budget was silently wasted on dead code for an entire solver version.

Version 3. After removing the broken cascade pull and adding pivot rotations, crankshaft, and rebridging, S9 improved from gap = 14 to gap = 2. The grid-based pivot optimization further improved throughput by 1.4–1.5 $\times$.

Sequence hallucination. The AI initially used incorrect sequences for S6 and S10 from parametric memory rather than verifying against the competition source. Runtime verification against the live JSON caught this before submission.

These failure modes—geometric impossibility of a proposed algorithm, silent dead code, hallucinated benchmark data—are characteristic of current LLMs and underscore the necessity of runtime verification.

5.1 Limitations and Future Work

Three improvements could close the remaining gaps: (1) Wang–Landau flat-histogram sampling [Wüst and Landau(2012)], which explicitly forces exploration of all energy levels; (2) incremental ΔE computation for pivot moves, amplifying the grid collision advantage from 1.5 $\times$ to potentially 10–50 $\times$ by eliminating the $O(n^2)$ recompute; (3) adaptive temperature spacing using exchange-rate feedback.

6 Conclusion

We presented a six-operator REMC solver for 2D HP protein folding and showed through controlled ablation that pivot rotations account for approximately 80% of the total improvement over local-move-only search. Grid-based pivot collision provides a measured $1.4\text{--}1.5\times$ throughput gain. On CAISc 2026, the solver matched S5 and achieved 212/225 total contacts (94.2%). The AI development process documented here—including a geometrically impossible move operator, hallucinated sequences, and missing pivots in v1—provides concrete evidence of LLM failure modes and the essential role of experimental validation in AI-driven science.

The verification artifact (coordinate files for S5–S10) and the complete Jupyter notebook are submitted as supplementary material.

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Conference For AI Scientists 2026 - AI Involvement Checklist

Research Stage Assessment

1. Hypothesis development:

Answer: [D]

Iteration Effort: [Medium]

Explanation: The AI system selected the HP protein folding problem from five verifiable options by analyzing feasibility, search-space structure, and verification complexity. It reviewed the literature spanning Lesh (2003), Thachuk (2007), Wüst & Landau (2012), and Khandoker (2025), identifying the move-operator design space and REMC as the most viable framework. The human provided the conference call for papers and compute constraints.

230 **2. Experimental design and implementation:**
 231 Answer: [D]
 232 Iteration Effort: [High]
 233 Explanation: The AI designed and implemented the solver in Python/Numba across three
 234 major versions: (v1) SA-only without pivots, (v2) REMC with pivots and a broken cascade
 235 pull, (v3) removal of the geometrically impossible cascade pull, addition of crankshaft
 236 and rebridging, and the grid-based pivot collision optimization. Over 100 substantive code
 237 iterations were performed, including debugging a silent-failure cascade-pull bug. The AI
 238 also designed and executed all four ablation experiments.

239 **3. Analysis of data and interpretation of results:**
 240 Answer: [D]
 241 Iteration Effort: [Medium]
 242 Explanation: The AI analyzed checkpoint JSONs, identified the $O(n \cdot s)$ pivot bottleneck
 243 via complexity analysis, designed and ran ablation experiments, generated all figures, and
 244 interpreted results including the cascade-pull geometric impossibility proof. The human
 245 uploaded raw outputs and conformation images.

246 **4. Writing:**
 247 Answer: [D]
 248 Iteration Effort: [Medium]
 249 Explanation: The AI wrote the complete paper including all sections, algorithm pseudocode,
 250 tables, figure captions, references, and both checklists. The human provided the $\text{\LaTeX}$
 251 template.

252 **AI System Documentation**

253 **5. AI systems used:**
 254 Description: A frontier large language model (LLM) served as primary scientific author for
 255 all stages of the research process: problem selection, literature review, algorithm design,
 256 Python/Numba implementation, iterative debugging, ablation experiment design, perfor-
 257 mance optimization, result analysis, and paper composition. The LLM operated in interactive
 258 conversational mode with a human supervisor who provided compute resources (Kaggle T4
 259 GPUs, Lightning AI A100) and high-level direction. The solver itself is a classical REMC
 260 algorithm; no neural networks, reinforcement learning agents, or learned components were
 261 used in the search procedure.

262 **6. Observed AI Limitations:**
 263 Description: Three significant failure modes were observed during development: (1) The AI
 264 implemented a cascade-pull move whose geometric precondition is impossible on the 2D
 265 square lattice—two bonded lattice positions share no common neighbor, so the precondition
 266 was never satisfied. This dead code silently consumed 25% of the move budget for an entire
 267 solver version. (2) The AI initially used incorrect benchmark sequences for S6 and S10
 268 from parametric memory, producing spurious improvement claims. Runtime verification
 269 against the competition JSON caught this error. (3) The first solver version lacked pivot
 270 moves entirely, despite their prominence in the cited literature, producing results far below
 271 state of the art. The importance of pivots was only recognized after analyzing poor initial
 272 results.

273 **Conference For AI Scientists 2026 - Reproducibility and Responsibility** 274 **Checklist**

275 **1. Claims**
 276 Question: Do the main claims made in the abstract and introduction accurately reflect the
 277 paper’s contributions and scope?
 278 Answer: [Yes]

279 Justification: The abstract states we matched S5 and reached 94.2% of contacts across
 280 S5–S10; Table 2 confirms this. The ablation claim is supported by Table 1 and Figure 1.
 281 The pivot speedup is measured in Figure 2.

282 **2. Limitations**

283 Question: Does the paper discuss the limitations of the work performed by the AI system(s)?

284 Answer: [Yes]

285 Justification: Section 5 discusses three specific AI failure modes. Section 5.1 discusses
 286 remaining gaps, S10 under-allocation, and high-H-density difficulty.

287 **3. Theory assumptions and proofs**

288 Question: For each theoretical result, does the paper include the full set of assumptions and
 289 a complete (or correct) proof?

290 Answer: [NA]

291 Justification: No new theorems are proved. NP-completeness is cited from prior work. The
 292 pivot collision correctness argument is informal.

293 **4. Experimental result reproducibility**

294 Question: Does the paper fully disclose all the information needed to reproduce the main
 295 experimental results of the paper?

296 Answer: [Yes]

297 Justification: Full source code (Jupyter notebook) is submitted as supplementary material.
 298 All hyperparameters are specified in Section 3. Benchmark sequences are verified at runtime
 299 against the live competition JSON.

300 **5. Open access to data and code**

301 Question: Does the paper provide open access to the data and code?

302 Answer: [Yes]

303 Justification: The complete Jupyter notebook, all checkpoint JSONs, and submission coordi-
 304 nate files for S5–S10 are provided as supplementary material on OpenReview.

305 **6. Experimental setting/details**

306 Question: Does the paper specify all the training and test details necessary to understand the
 307 results?

308 Answer: [Yes]

309 Justification: Section 4.1 specifies the hardware platform (Lightning AI A100-40GB, 4
 310 CPUs), wall-clock budget (10 hours), and software stack (Python 3, Numba JIT). Section 3
 311 details all hyperparameters including temperature bounds, move probabilities, replica counts,
 312 and compute scaling rules.

313 **7. Experiment statistical significance**

314 Question: Does the paper report error bars suitably and correctly?

315 Answer: [No]

316 Justification: The benchmark results (Table 2) are from a single 10-hour run. The ablation
 317 study (Table 1) reports both best and mean over 10 independent seeds. Full variance analysis
 318 across multiple long runs was not feasible within the competition deadline.

319 **8. Experiments compute resources**

320 Question: For each experiment, does the paper provide sufficient information on the com-
 321 puter resources needed to reproduce them?

322 Answer: [Yes]

323 Justification: Lightning AI A100-40GB instance, 4 CPU cores, 10-hour wall clock. Per-
 324 sequence compute times are reported in Table 2. Ablation experiments specify step counts
 325 and seed counts.

326 **9. Research Ethics**

327 Question: Does the research conducted in the paper conform, in every respect, with ethical
328 norms?

329 Answer: [Yes]

330 Justification: This work solves a well-known combinatorial optimization benchmark. No
331 human subjects, sensitive data, or dual-use concerns are involved.

332 10. **Broader impacts**

333 Question: Does the paper discuss both potential positive and negative societal impacts of
334 the work performed?

335 Answer: [NA]

336 Justification: The HP model is a simplified abstraction used for algorithmic research. This
337 work does not produce real protein structures and has no direct societal impact, positive or
338 negative.